

NED UNIVERSITY OF ENGINEERING AND TECHNOLOGY



PROJECT REPORT ON

**Automated Extraction and Exploratory Analysis of
Court Judgment Data Project Carried Out**

BY

NAME: SYED MOHAMMAD HUZAIFA

ROLL No: CTAI-022 (2023 Batch)

Under The Supervision Of

DR.WASEEMULLAH NAZIR

Associate Professor

Department Of Computer Science and Information Technology

ABSTRACT

Legal judgment documents contain rich information critical for understanding judicial trends, yet their highly unstructured format makes large-scale analysis difficult. This project focuses on transforming a collection of court judgment PDFs into a structured, machine-readable dataset to enable systematic exploration of legal patterns. Using a custom extraction pipeline, key metadata—including case details, parties involved, judgment dates, legal sections, and outcomes—were parsed, cleaned, and standardized into a unified dataset. Exploratory Data Analysis (EDA) was conducted to uncover initial trends, revealing dominant judgment outcomes, frequently cited legal sections, and substantial variation in judgment length across cases. Early findings suggest that data inconsistencies, date irregularities, and noisy textual fields are significant challenges in legal document processing but can be effectively addressed through tailored preprocessing and feature engineering. This report summarizes progress up to the EDA stage, with future work focused on advanced modeling and deeper text-based legal analytics to derive actionable insights from judicial data.

Contents

Under The Supervision Of	1
ABSTRACT	2
1. INTRODUCTION	4
1.1 Background.....	4
1.2 Problem Statement	4
1.3 Objectives	4
2. DATA COLLECTION & UNDERSTANDING	5
Columns Extracted	5
2.1 Data Preprocessing.....	6
2.1.1 Data Quality Assessment (Issues & Solutions)	6
B. Cleaning Steps Applied (Short Version).....	6
3. FEATURE ENGINEERING	6
3.1 Data Standardization and Transformation	7
Feature 1: Date Standardization (Year and Month)	7
Feature 2: Text Length Features (Word Count and Character Count)	7
3.2 Engineered Features	7
Feature 3: Section Count (Legal Complexity Indicator)	7
Feature 4: Primary Legal Section Extraction.....	7
Feature 5: Numerical Summary Features (Word & Character Count Statistics)	8
4. EXPLORATORY DATA ANALYSIS (EDA) INSIGHTS	8
4.1 Data Quality Assessment (Missing Values Analysis)	9
Fig 4.0 (A): Missing Values Heatmap / Table	9
4.1. Categorical Feature Analysis (Outcome, Year, and Legal Sections)	9
A. Judgment Outcome Distribution	9
B. Judgment Year Distribution	10
C. Month-Wise Case Distribution.....	11
4.2. Numerical and Textual Distribution Analysis (Judgment Length & Complexity)	12
D. Word Count Distribution	13
E. Average Judgment Length by Outcome	13
F. Character Count Distribution.....	15
G. Outlier Analysis	15
4.3. Correlation Analysis.....	16
H. Correlation Matrix / Heatmap	17
5. TOOLS & TECHNOLOGIES USED	17
6. FUTURE WORK.....	18
7. CONCLUSION.....	18

1. INTRODUCTION

1.1 Background

The legal system generates vast amounts of information contained within unstructured court judgment PDFs. This data, characterized by varying formats, inconsistent writing styles, and complex legal jargon, presents a significant challenge for quantitative analysis. This project aims to unlock the analytical potential of this data by transforming it into a structured, machine-readable format.

1.2 Problem Statement

The primary challenge is the inability to perform large-scale, systematic analysis of court cases due to the unstructured, noisy, and inconsistent nature of the original PDF documents. A dedicated process is required to extract, clean, and structure the key legal metadata before any meaningful insights can be derived.

1.3 Objectives

The main objectives of this phase were:

1. To design and execute a custom pipeline to **extract essential metadata** from unstructured PDF court judgments.
 2. To perform comprehensive **data cleaning and standardization** to create a reliable, analysis-ready dataset.
 3. To conduct **Exploratory Data Analysis (EDA)** to understand the data's distributions, structures, and preliminary trends.
-

2. DATA COLLECTION & UNDERSTANDING

The dataset for this project was compiled directly from court judgment PDF files, making the data collection phase significantly more challenging than working with pre-structured, publicly available datasets. Court judgments are long, unstructured documents characterized by inconsistent formatting, varying writing styles, and embedded legal references scattered throughout narrative text. Locating a source that provided such judgments in bulk was itself difficult, as very few openly available repositories offer machine-readable legal data. Once the documents were obtained, converting them into a structured dataset required extensive effort and customization.

To accomplish this, a dedicated Python script (**extract_judgments.py**) was developed to automate the extraction process. The script integrated several libraries to handle the complexity of unstructured legal text:

- **pdfplumber** → used to extract raw text from each page of the PDF judgments.
- **Regular Expressions (Regex)** → designed to detect and extract legal metadata such as case numbers, dates, parties, advocates, judges, cited sections, and outcomes from within the free-form text.
- **Dateutil** → used to standardize the many inconsistent date formats encountered in the documents.
- **pandas** → used to organize the extracted information into a structured CSV file suitable for cleaning and analysis.

The output of this pipeline was a raw CSV file (**out.csv**) containing metadata and selected text segments from each judgment. A total of **881 records** were collected and prepared for further processing.

Columns Extracted

The extraction script captured a wide range of metadata, including:

- **filename** → name of the original judgment PDF.
- **judge_folder** → folder representing the judge or bench associated with the case.
- **case_no** → unique identifier assigned to each case.
- **case_type** → case category (e.g., Civil, Criminal, Petition).
- **date** → date of the final judgment.
- **hearing_date** → date of the hearing preceding the judgment.
- **petitioner / respondent** → names of the involved parties.
- **advocates_petitioner / advocates_respondent** → legal counsel representing each party.
- **judges, bench_size, judgment_author** → composition of the judicial panel and author of the judgment.
- **originating_court, originating_case_no** → references to the lower court and original case number.
- **constitutional_refs, precedents, sections** → citations of constitutional articles, referenced prior cases, and legal sections applied.

2.1 Data Preprocessing

2.1.1 Data Quality Assessment (Issues & Solutions)

The raw dataset extracted from the judgment PDFs was highly inconsistent and required extensive preprocessing before analysis. The unstructured nature of the text introduced several data quality issues, including missing fields, inconsistent naming patterns, irregular date formats, and categorical noise. The table below summarizes the major problems identified and the solutions implemented.

Issue	Problem in Raw Data	Solution Implemented
Missing Values	Missing case_no and other essential fields.	Dropped essential-missing records; filled non-critical fields with "Unknown".
Text Inconsistency	Irregular party/judge/advocate names.	Standardized text (capitalization, trimming); removed redundant labels.
Date Format Errors	Multiple inconsistent date formats.	Converted to ISO (YYYY-MM-DD) using Dateutil.
Categorical Noise	Irregular outcome and section labels.	Normalized outcomes; extracted sections into structured list.
Duplicates	Same case appearing more than once.	Removed duplicates using filename + case_no.

B. Cleaning Steps Applied (Short Version)

Cleaning was performed in **analysis.ipynb**:

- Handled Missing Values**
 - Dropped essential-missing rows
 - Filled non-critical fields with "Unknown"
 - Standardized Text Columns**
 - Removed labels (Petitioners, Unknown)
 - Capitalized + trimmed names
 - Standardized Dates**
 - Converted date and hearing_date to ISO format
 - Fixed parsing errors
 - Removed Duplicates**
 - Cleaned Judges & Advocates**
 - Merged duplicate name formats
 - Standardized advocate lists
 - Dropped Unneeded Long Text Fields**
 - Removed petition_text
 - Kept word_count + char_count
 - Standardized Outcomes & Sections**
 - Normalized outcome labels
 - Extracted sections into lists
-

3. FEATURE ENGINEERING

Feature engineering was carried out to extract meaningful attributes from raw textual, temporal, and categorical fields. These engineered features help in identifying trends, understanding case complexity, and preparing data for further analysis.

3.1 Data Standardization and Transformation

Feature 1: Date Standardization (Year and Month)

Action Taken:

- The judgment date was cleaned and standardized.
- Two numerical time-based features were extracted: **year** and **month**.

Rationale:

Breaking the date into year and month allows for temporal trend analysis, including yearly caseload patterns and seasonal fluctuations.

Feature 2: Text Length Features (Word Count and Character Count)

Action Taken:

- The total number of words and characters in each judgment document was calculated.
- Their distributions were analyzed to identify variations in judgment length and writing patterns.

Rationale:

Judgments with higher text length often indicate more complex cases, detailed arguments, or extensive legal reasoning. These numerical features help detect outliers and trends in case complexity.

3.2 Engineered Features

Feature 3: Section Count (Legal Complexity Indicator)

Action Taken:

- The list of legal sections referenced in each case was parsed and standardized.
- The number of sections in each case was counted as a measure of legal complexity.

Rationale:

Cases involving multiple legal sections tend to cover broader issues and may require more detailed judicial evaluation. Section count serves as a useful indicator of complexity.

Feature 4: Primary Legal Section Extraction

Action Taken:

- From the standardized section list, the first listed section was extracted to represent the primary legal basis of the case.

Rationale:

The primary section helps classify cases into broad legal categories, supporting frequency analysis and pattern discovery.

Feature 5: Numerical Summary Features (Word & Character Count Statistics)

Action Taken:

- Statistical summaries such as averages, ranges, and distributions of word and character counts were generated and studied.

Rationale:

These summaries provide insight into the overall structure and verbosity of judgments, assisting in identifying unusually short or long document

4. EXPLORATORY DATA ANALYSIS (EDA) INSIGHTS

The Exploratory Data Analysis (EDA) phase was conducted to understand the statistical properties, temporal patterns, legal complexity indicators, and textual characteristics of the

court judgments dataset. Multiple visual and numerical analyses were used to uncover underlying trends, highlight data issues, and inform the subsequent modelling strategy. The insights in this section are categorized into three primary dimensions: categorical patterns, numerical/textual trends, and correlation structures.

4.1 Data Quality Assessment (Missing Values Analysis)

A foundational step in EDA is assessing the integrity of the dataset by identifying missing or incomplete values.

Fig 4.0 (A): Missing Values Heatmap / Table

Insight:

- The *outcome* and *judgment text* fields contain notable missing values, indicating incomplete metadata in several records.
- Some date components (e.g., filing year, month) also contain sparse entries.
- These gaps justify necessary preprocessing such as imputation, exclusion, or flag encoding.
- Missing textual fields may affect downstream NLP tasks and must be handled with care.

This assessment validates the need for cleaning procedures and ensures transparency in dataset reliability.

4.1. Categorical Feature Analysis (Outcome, Year, and Legal Sections)

This subsection examines the key categorical variables that define judgment outcomes, filing timelines, and cited legal sections. Each visualization is followed by a detailed interpretation.

A. Judgment Outcome Distribution

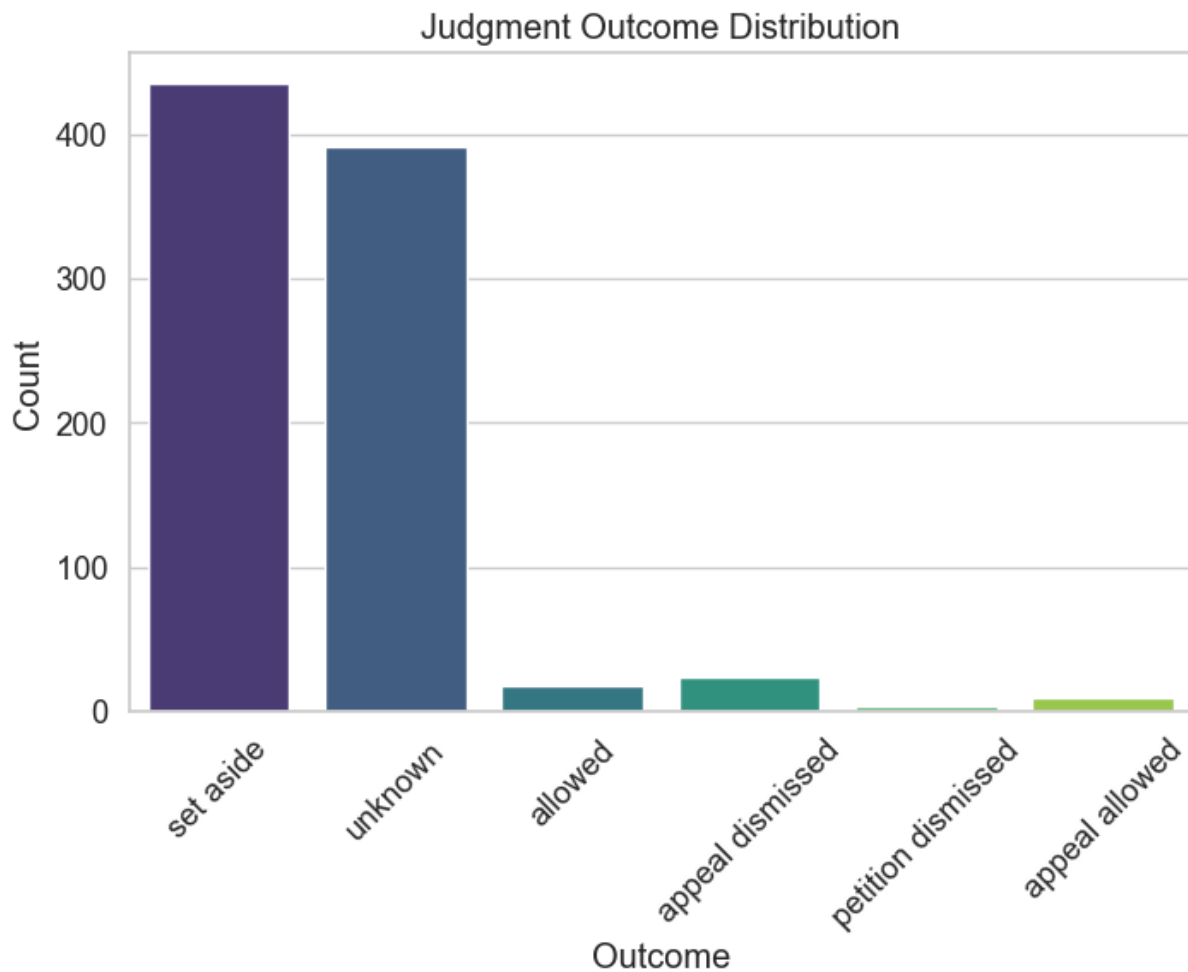


Fig 4.1 (A): Outcome Countplot

Insight:

The dataset exhibits a strong imbalance in judgment outcomes.

- *Set aside* is the dominant category, representing the largest proportion of cases.
- A substantial number of cases fall into the *unknown* category, indicating incomplete entries or missing metadata.
- *Appeal dismissed* cases form a much smaller cluster.

The imbalance in outcome frequencies suggests that certain judgment types are underrepresented, which may limit predictive modelling unless appropriate resampling strategies are considered.

B. Judgment Year Distribution

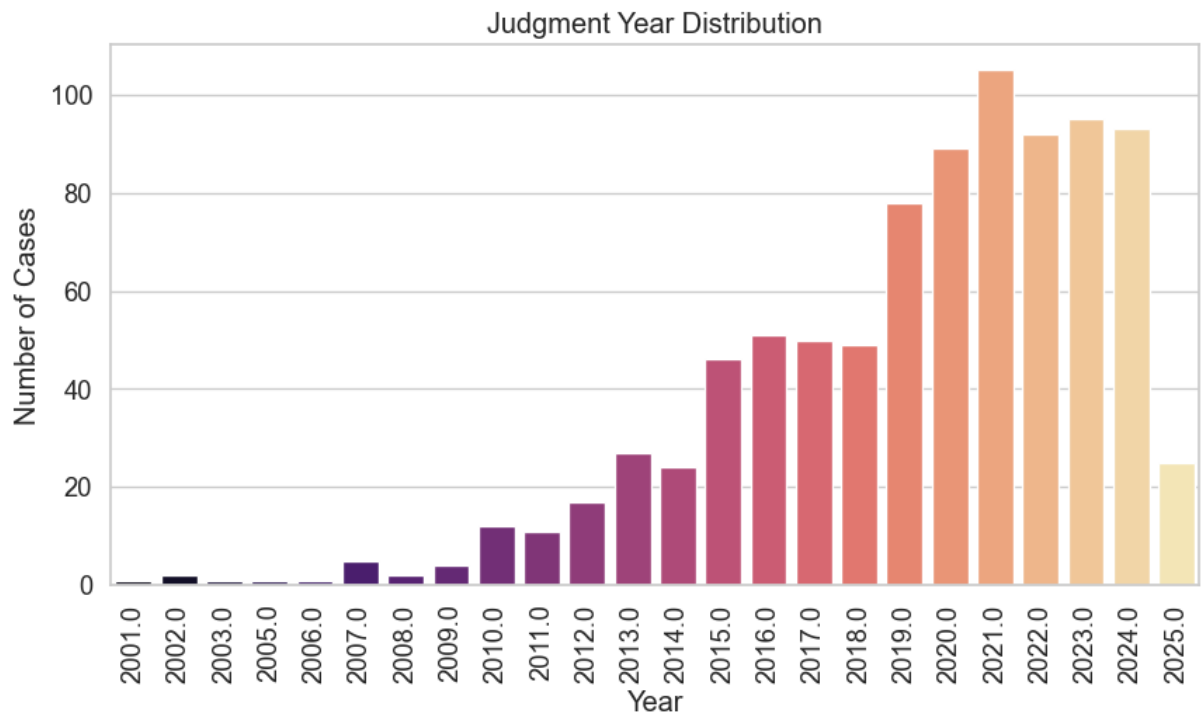


Fig 4.1 (B): Year Countplot

Insight:

The distribution of cases across years is uneven and shows multiple gaps.

- A large portion of years contain few or no judgments.
- This sparsity is likely due to incomplete archival data or variation in digital case availability across different time periods.
- The uneven yearly distribution restricts year-wise temporal modelling but still supports cluster-level trend analysis.

C. Month-Wise Case Distribution

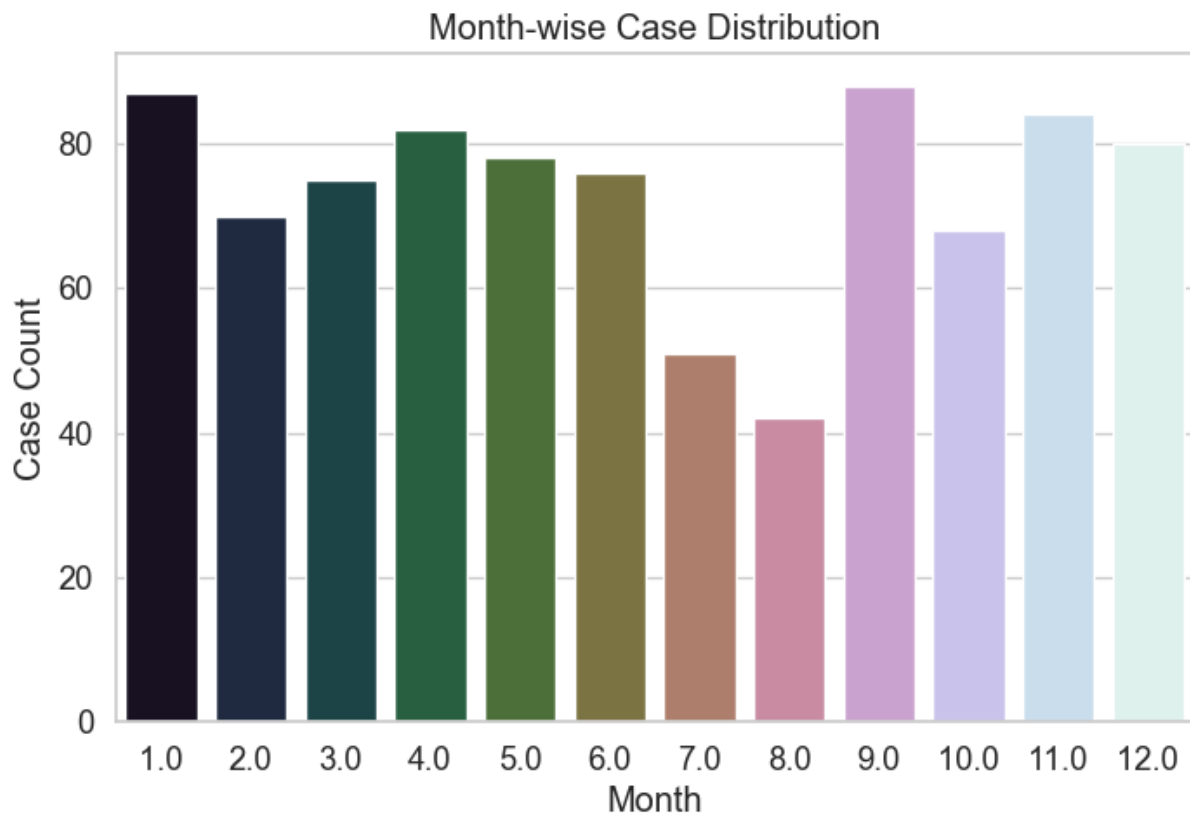


Fig 4.1 (C): Month Countplot

Insight:

A mild seasonality pattern emerges in monthly case counts, though the differences are not extreme.

- Some months show higher case volumes; however, the variation is likely influenced by inconsistent annual data rather than true seasonality.
- The pattern helps identify operational case load tendencies across the calendar cycle.

4.2. Numerical and Textual Distribution Analysis (Judgment Length &

Complexity)

The numerical analysis focuses on text-based features—particularly word count and character count—to understand the variation in judgment length and document verbosity.

D. Word Count Distribution

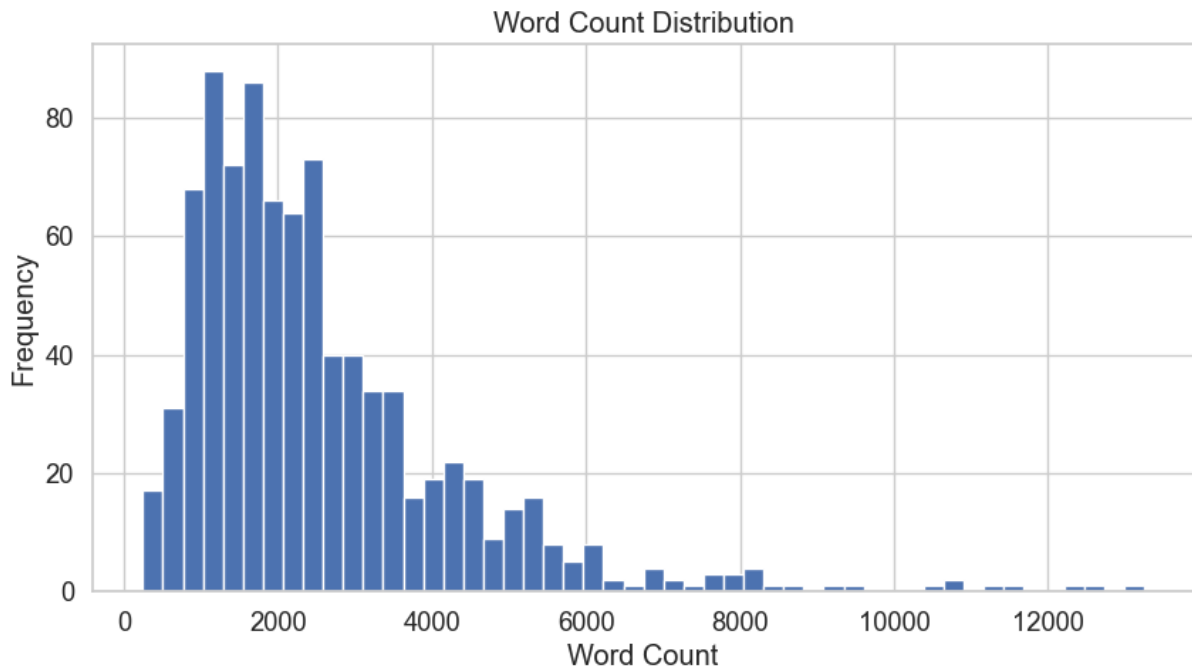


Fig 4.2 (A): Word Count Histogram

Insight:

- Judgment word counts display a right-skewed distribution.
 - A few judgments exceed **13,000 words**, representing exceptionally detailed or complex cases.
 - The mean word count (~2,559) is significantly higher than the median (~2,138), confirming the influence of long-document outliers.
 - The long tail indicates that some cases require extensive justification, procedural history, and legal interpretation.
-

E. Average Judgment Length by Outcome

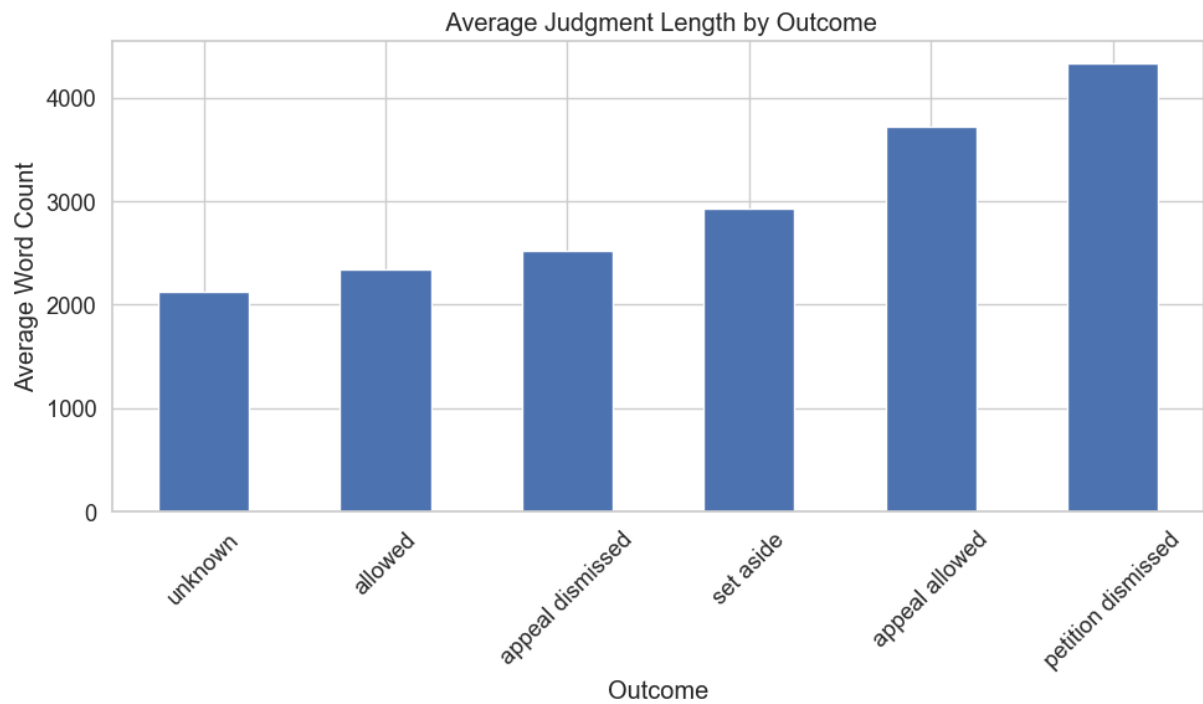


Fig 4.2 (B): Bar Chart — Mean Word Count per Outcome

Insight:

- Certain outcomes (e.g., *Set Aside* or *Dismissed*) tend to have longer judgments, reflecting detailed reasoning or procedural history.
- *Unknown* outcomes exhibit varying lengths, likely due to metadata gaps.
- This comparison offers an early indication of whether outcome types differ in complexity.

F. Character Count Distribution

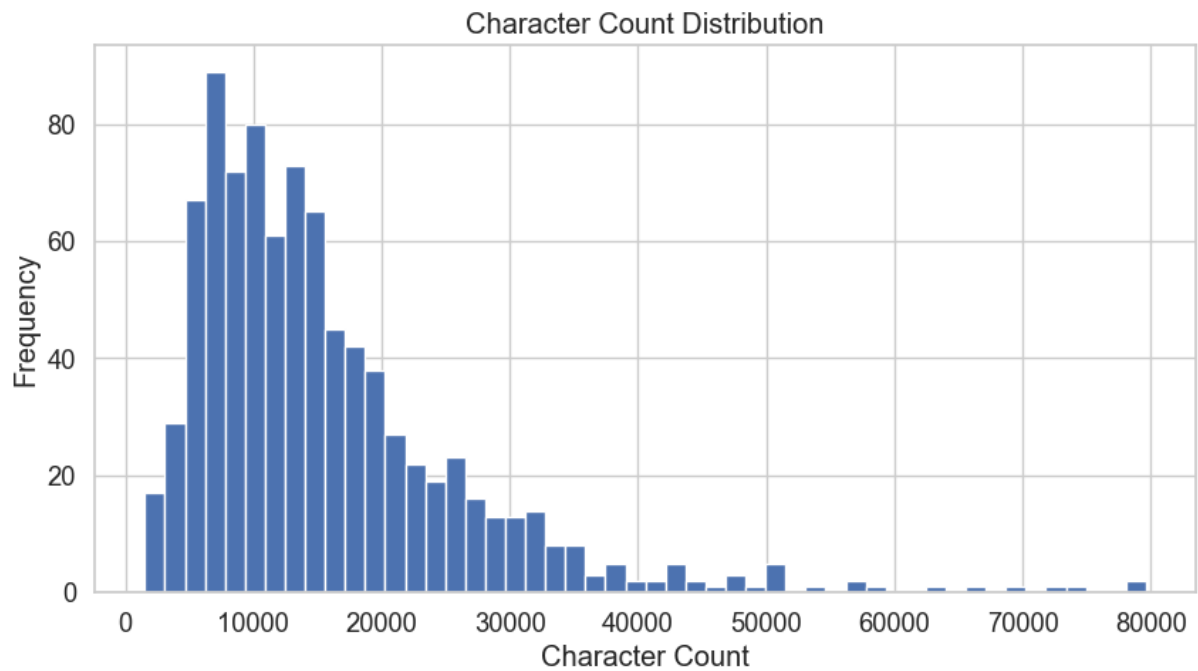


Fig 4.2 (C): Character Count Histogram

Insight:

- Character counts show an even more pronounced right skew.
 - Maximum character count exceeds **79,000**, consistent with the longest judgments from the word count analysis.
 - This feature is important for detecting abnormally short or unusually long judgment documents, which may require separate handling in preprocessing.
-

G. Outlier Analysis

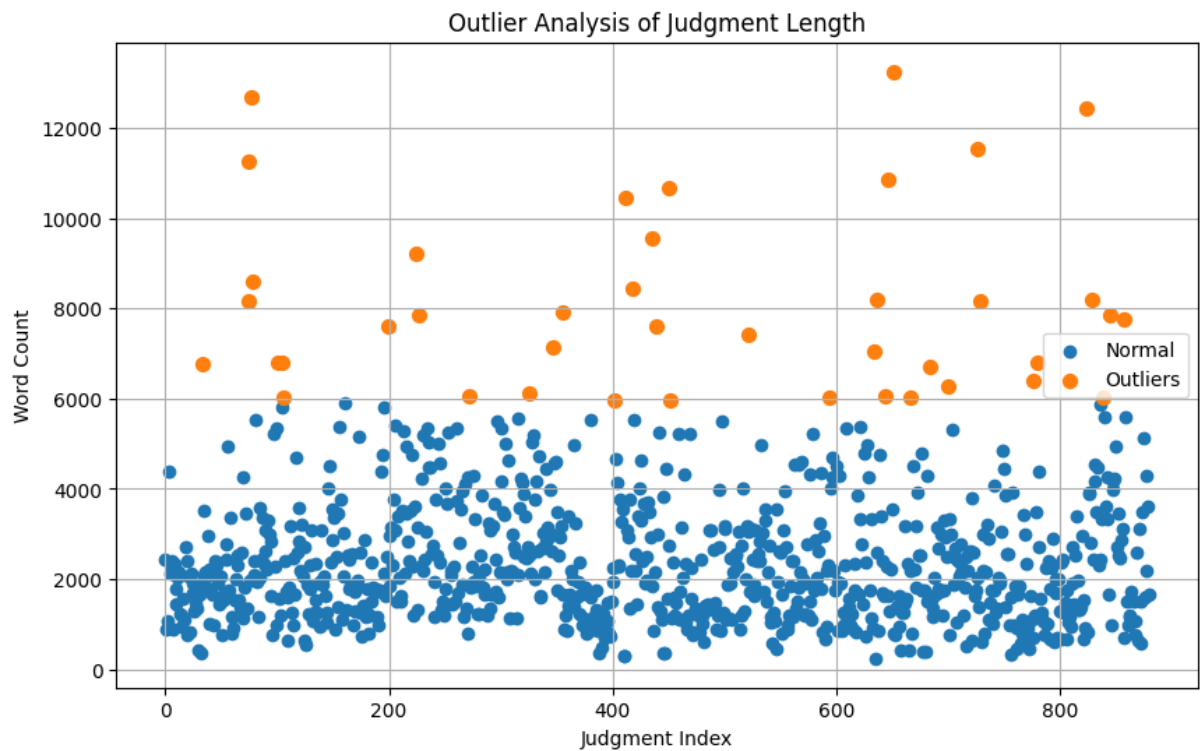


Fig 4.2 (D): Outlier Detection Plot

Insight:

- The longest judgments show exceptionally high word and character counts, typically representing multi-hearing or appellate-level cases.
- The shortest judgments contain very minimal text, often due to clerical orders, metadata gaps, or preprocessing artifacts.
- These outliers reveal significant variation in judgment complexity and guide preprocessing steps such as normalization or length-based filtering..

4.3. Correlation Analysis

H. Correlation Matrix / Heatmap

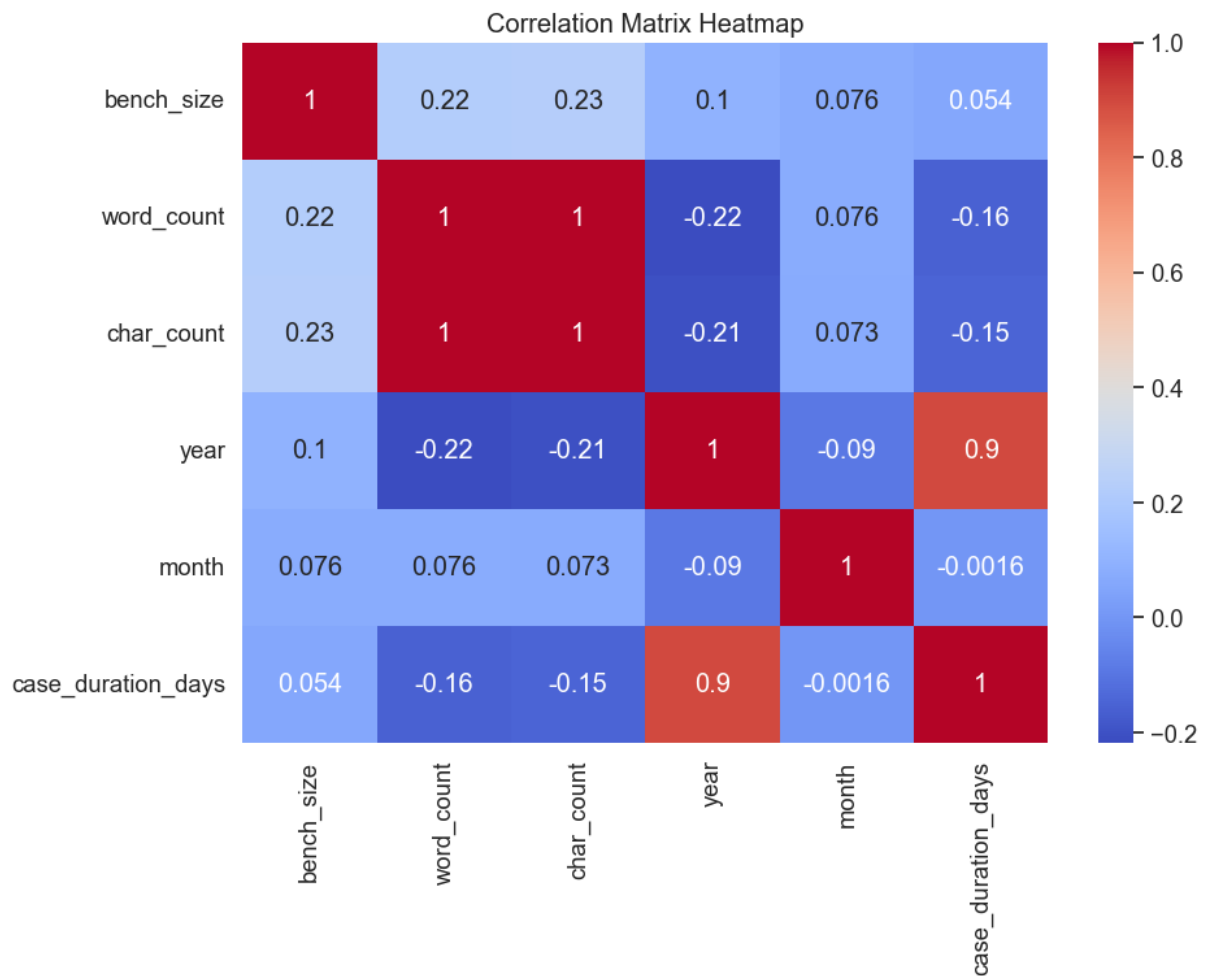


Fig 4.3 (A): Correlation Heatmap

Insight:

- Weak correlation between judgment outcomes and textual length features suggests that the outcome type does not strongly depend on document verbosity.
- Character count and word count exhibit a very strong positive correlation, as expected for textual data.
- No strong multicollinearity appears among other engineered features, which is advantageous for downstream modeling.

5. TOOLS & TECHNOLOGIES USED

The following tools and technologies were instrumental in the data pipeline:

- **Python:** The core scripting environment.
 - **pdfplumber:** Used for robust extraction of raw text from PDF files.
 - **Regex (Regular Expressions):** Critical for parsing and capturing specific legal metadata fields (e.g., case numbers) from unstructured text.
 - **pandas:** Used for data manipulation, cleaning, and structuring the data into a usable CSV format.
 - **Dateutil:** Utilized for handling and standardizing the diverse date formats encountered in the legal documents.
 - **matplotlib** and **seaborn:** Libraries employed for data visualization during the EDA phase.
-

6. FUTURE WORK

Moving forward, the project will transition from data preparation to modeling and deeper analysis:

1. **Modeling Data Preparation:** Implement further feature engineering, including **binary encoding** (for categorical features like `case_type`) and **data standardization/scaling** (for numerical features like `word_count`) in preparation for predictive models.
 2. **Predictive Modeling:** Develop machine learning models to predict the judgment outcome based on structured metadata features.
 3. **In-Depth Text Analysis:** Perform detailed analysis on the cited precedents and the frequency of individual legal sections from the `sections_list`.
-

7. CONCLUSION

This project successfully overcame the fundamental challenge of unstructured legal data by implementing a robust data extraction and cleaning pipeline. The process transformed raw judgment PDFs into a high-quality dataset of 881 cases. EDA provided crucial initial insights, confirming the prevalence of "**set aside**" judgments and establishing the wide range of document complexity as measured by **word count**. This structured dataset is now ready for advanced feature engineering and machine learning applications in the domain of legal analytics